Pranav M V
192411030
CO - 2 AT- 2
Question - 2

Flow 2

**Comparison of DSDV and AODV Routing Protocols in Mobile Ad Hoc Networks (MANETs)**

**Introduction:**
In a mobile network, devices may move from one location to another. Therefore, the wireless connections between them can change frequently.
A routing protocol is a set of rules that helps devices determine how to forward data from a source to a destination.

**Two commonly studied MANET routing protocols are:**
- DSDV (Destination-Sequenced Distance-Vector): A proactive routing protocol that maintains routes before they are needed.
- AODV (Ad hoc On-Demand Distance Vector): A reactive routing protocol that discovers routes when they are needed.

The main difference is that DSDV maintains routing information in advance, while AODV discovers routes on demand.

**Network Topology Diagram**




**Network Topology Diagram**

**Imagine four mobile devices connected in a wireless network:**
- Node A: Source device that sends the data.
- Node B: Intermediate node that forwards the data.
- Node C: Alternative intermediate node.
- Node D: Destination device that receives the data.

**The main route is:**

If the link between A and B or between B and D fails, the network may use an alternative route, such as:
Whether this alternative route can be used depends on whether the required links exist and the routing protocol has discovered or maintained a valid route.

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